

Box–Cox λ as a Redshift-Resolved Shape Statistic for Galaxy Colour Distributions: Advantages over Johnson S_U Parameterisation

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Working draft — target: MNRAS or A&A Methods

Abstract

We argue that the Box–Cox power parameter λ is a superior redshift-resolved shape statistic for galaxy colour distributions compared to the two shape parameters (a , b) of the Johnson S_U family. Although the Johnson S_U distribution is flexible, its shape parameters are strongly correlated under maximum-likelihood optimisation, producing numerical instability when tracking smoothly evolving distributions across redshift bins. Box–Cox λ is a single interpretable parameter — the power that best normalises a positive distribution — with direct physical meaning: $\lambda = 1$ implies additive (Normal) scatter, $\lambda \rightarrow 0$ implies multiplicative (log-Normal) scatter, and $\lambda < 0$ implies a distribution sharply bounded on one side, consistent with an absorbing state such as a consolidated red sequence. We demonstrate using $u-r$ colour distributions for 165,488 GAMA DR4 galaxies in seven redshift bins ($z = 0.002-0.35$). Box–Cox λ increases monotonically from -0.20 to $+2.70$ (Spearman $r = 1.000$, $p < 0.001$) while JSU parameter a jumps non-monotonically between -11.7 and $+20.7$ across the same bins. A bootstrap null test confirms the λ trend exceeds sampling noise at every bin.

1. Introduction

Parametric fitting of galaxy property distributions is common for characterising populations and tracking redshift evolution. For galaxy colours, the $u-r$ distribution is bimodal at low redshift but its shape evolves with cosmic time. A useful shape statistic must be: (i) sensitive to the full distributional form; (ii) numerically stable across bins of varying size; and (iii) physically interpretable.

The Johnson S_U distribution (Johnson 1949) is widely used in finance and engineering. Its shape parameters a and b control skewness and kurtosis but are strongly correlated under maximum-likelihood optimisation, causing the optimiser to find numerically distant solutions for statistically equivalent data.

The Box–Cox power transform (Box & Cox 1964) provides a single-parameter alternative with a unimodal, well-conditioned likelihood surface and direct physical interpretation. This paper makes the case for Box–Cox λ as a standard diagnostic in redshift-resolved distribution analysis.

2. Statistical Background

2.1 Johnson S_U Distribution

The JSU family is defined by $z = a + b \sinh^{-1}((x - \xi)/\text{scale})$, $z \sim N(0,1)$. When the true distribution is close to symmetric, many (a , b) pairs fit equally well, and small perturbations in data cause the optimiser to jump between equivalent solutions — the artefact demonstrated in Section 4.

2.2 Box–Cox Power Transform

$y(\lambda) = (x^\lambda - 1)/\lambda$ for $\lambda \neq 0$; $y(0) = \log(x)$. Estimated by maximising the profile log-likelihood of $y \sim N(\mu, \sigma^2)$. The surface is smooth and unimodal for $N \gtrsim 100$. Physical interpretation: $\lambda \approx 1 \rightarrow$ Normal (additive scatter); $\lambda \rightarrow 0 \rightarrow$ log-Normal (multiplicative); $\lambda < 0 \rightarrow$ sharply bounded, one-way absorbing boundary.

2.3 Bootstrap Null Test

500 subsamples of the same size as each redshift bin, drawn with replacement from the full galaxy pool (ignoring redshift). The 2.5th–97.5th percentiles of the resulting λ distribution form the 95% null envelope. Monotonic trend assessed with Spearman r and Kendall τ .

3. Data

GAMA DR4 StellarMassesLambdarv24 with quality cuts $0.5 < u-r < 4.0$, $\log M^* > 8.0$, $0.002 < z < 0.35$: 165,488 galaxies in seven bins ($\Delta z = 0.05$; $N = 3,900\text{--}34,000$). GAMA is $>98\%$ spectroscopically complete to $r < 19.8$.

4. Results

4.1 JSU Instability Demonstrated

Table 1 shows all shape parameters by redshift bin. Skewness decreases smoothly from +0.90 to -0.82 . Box–Cox λ increases smoothly from -0.20 to $+2.70$. JSU parameter a , however, jumps from -2.76 to -11.68 to $+20.68$ across the first three bins — a factor of >30 while skewness changes by only 0.75. Figure 1 shows the contrast directly.

Table 1. Shape parameters by redshift bin (GAMA DR4, $N = 165,488$)

z	N	JSU a	JSU b	Box–Cox λ	Skewness	Kurt
0.026	3,932	-2.763	+2.210	-0.200	+0.899	+0.824
0.075	16,579	-11.677	+5.925	+0.431	+0.263	-0.750
0.125	29,589	+20.679	+7.171	+1.125	-0.148	-1.046
0.175	32,999	+9.776	+3.566	+1.694	-0.387	-0.897
0.225	27,116	+3.431	+2.112	+1.995	-0.523	-0.623
0.275	33,685	+2.802	+1.752	+2.532	-0.745	-0.188
0.325	21,588	+2.836	+1.802	+2.701	-0.821	+0.069

JSU a : Spearman $r = +0.29$, $p = 0.53$ (not significant). Box–Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU a varies by $>30\times$ while skewness changes by only 0.75 — instability is numerical, not physical.

Figure 1. Three-panel comparison. Left: JSU parameter a vs redshift — large non-monotonic jumps despite smooth underlying evolution. Centre: Box–Cox λ with 95% bootstrap null envelope — smooth monotonic increase. Right: Skewness with null envelope — equally smooth. Red stars mark bins outside the null.

4.2 Box–Cox λ Tracks Physical Evolution

Figure 2 shows λ and skewness against redshift with 95% bootstrap null envelopes. Both parameters lie outside the null at every bin. Spearman $r = +1.000$ ($p = 4.0 \times 10^{-4}$) for λ ; -1.000 ($p = 4.0 \times 10^{-4}$) for skewness. The transition through $\lambda = 0$ near $z \approx 0.08$ may mark the epoch at which the red sequence first becomes the dominant population — a testable prediction for deeper surveys.

Figure 2. Box–Cox λ (left) and skewness (right) vs redshift with 95% bootstrap null envelopes. Both show perfect monotonic trends ($|Spearman\ r| = 1.00$, $p = 4 \times 10^{-4}$). All bins lie outside the sampling-noise null.

5. Discussion

5.1 When to Use Box–Cox λ vs JSU

JSU remains useful when a closed-form PDF is required (e.g. generating synthetic realisations or computing tail probabilities). For tracking distributional evolution with a scalar statistic, Box–Cox λ is preferable: single-parameter, well-conditioned, and physically interpretable.

5.2 Interpreting the λ Trend: Evolution vs Selection

A natural expectation is that galaxy colour distributions should be more Gaussian at higher redshift, when less quenching has occurred and the population is more uniformly star-forming. However, the observed λ trend — increasing from -0.20 at $z \approx 0.03$ to $+2.70$ at $z \approx 0.33$ — appears to contradict this. The resolution lies in two points.

First, $\lambda = 1$ marks the Normal distribution, not the upper end of the scale. The GAMA bins pass through $\lambda \approx 1$ near $z \approx 0.125$, which is the most Gaussian bin. The trend is a smooth transition from right-skewed (positive skewness, $\lambda < 1$) at low z to left-skewed (negative skewness, $\lambda > 1$) at high z — a shape flip, not a monotonic approach to or departure from Normality.

Second, GAMA is a flux-limited survey ($r < 19.8$). At low redshift the full galaxy population is visible, including blue star-forming dwarfs that dominate by number, producing a positively skewed $u-r$ distribution. At higher redshift, only the most luminous — and therefore most massive and reddest — galaxies satisfy the flux limit (Malmquist bias). Blue low-mass galaxies fall below the detection threshold, and the observable distribution becomes negatively skewed. The λ trend therefore reflects the interplay of genuine colour-distribution evolution and the magnitude-limited selection function, not pure cosmic evolution of a fixed population.

To isolate the evolutionary signal, the analysis should be repeated on a stellar-mass-complete subsample (e.g. $\log M^* > 10.5$ across all bins). In that case, one would compare the same class of galaxy at different epochs, and λ might plausibly evolve toward 1 at higher z if massive galaxies were more uniformly star-forming in the past. The DES application described in §5.3 offers a partial remedy: DES reaches $r \approx 24$, maintaining completeness to much lower masses at $z \lesssim 1$, which greatly reduces the Malmquist contamination of the shape trend.

5.3 Generalisability

The argument applies to any redshift-binned distribution analysis: stellar mass functions, SFR distributions, metallicity. Limitation: Box–Cox requires positive data; distributions spanning zero require a shift parameter. In such cases skewness is the most assumption-free scalar.

The approach is directly applicable to photometric surveys such as the Dark Energy Survey (DES), where the larger volume and depth ($z \lesssim 1.5$, ~ 300 million sources) would extend the evolutionary baseline by a factor of ~ 4 . The DES $g-r$, $r-i$, and $i-z$ colours are suitable inputs; the redMaGiC red-galaxy catalogue and redMaPPer cluster catalogue provide natural analogues of the GAMA group-environment analysis. One important modification is required: DES photometric redshift scatter ($\sigma_z/(1+z) \approx 0.02-0.05$) broadens the effective redshift bins and introduces inter-bin contamination. This uncertainty should be propagated into the bootstrap null test, either by drawing bootstrap subsamples from photo- z PDF realisations, or by adopting wider bins (e.g. $\Delta z = 0.1$) where scatter is sub-dominant. With these adjustments, a monotonic λ trend extending to $z \sim 1.5$ would constitute strong evidence for continuous colour-distribution evolution across more than half of cosmic time.

6. Conclusions

1. JSU parameters a and b show large non-monotonic variation across redshift bins due to parameter correlation on a near-flat likelihood ridge.
2. Box–Cox λ is a single, well-conditioned, physically interpretable shape statistic tracking the same evolution without numerical instability.
3. Applied to 165,488 GAMA galaxies, λ increases monotonically from -0.20 to $+2.70$ across $z = 0.03-0.33$ (Spearman $r = 1.000$, $p = 4 \times 10^{-4}$).
4. Bootstrap null testing confirms the trend exceeds sampling noise at every bin.

5. The observed λ trend in GAMA reflects both genuine colour evolution and Malmquist selection bias; a mass-complete subsample is required to isolate the purely evolutionary signal.
6. Box–Cox λ is recommended as the default shape statistic for redshift-resolved distribution analysis when a stable, interpretable scalar is required.
7. The framework extends naturally to deeper photometric surveys such as DES; photo- z scatter should be propagated into the bootstrap null, and mass-completeness limits should be checked before physical interpretation.

Data and Code Availability

GAMA DR4: <https://www.gama-survey.org/dr4/>. Python scripts available on request.

References

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